Single-Cell RNA-seq Analysis of Glioblastoma

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Introduction

Glioblastoma (GBM) is a highly heterogeneous brain tumor composed of multiple cellular populations with distinct transcriptional profiles. Single-cell RNA sequencing (scRNA-seq) enables the characterization of cellular heterogeneity at the level of individual cells and allows different cell populations to be identified and compared. The objective of this project was to perform a complete single-cell RNA-seq analysis of glioblastoma samples using a reproducible bioinformatics workflow. The analysis included sequencing quality control, read alignment and gene quantification, Seurat-based single-cell analysis, cell-type annotation, sample composition analysis, and exploratory copy-number variation (CNV) analysis using inferCNV.

Biological Question

Which cellular populations are present in glioblastoma samples, how do their proportions differ between samples, and can single-cell transcriptomic data be used to investigate large-scale copy-number variation patterns?

Dataset

The project uses publicly available single-cell RNA-seq data from three human glioblastoma samples.

Dataset information

- Organism: Homo sapiens
- Technology: Single-cell RNA sequencing
- Samples:
    - SRR10353960
    - SRR10353961

    - SRR10353962
- Total number of cells: 13,632
- Reference annotation: GENCODE v47
- Main analysis framework: Seurat
- CNV analysis: inferCNV

The three samples contained:

- SRR10353960 - 4,791 cells
- SRR10353961 - 4,652 cells
- SRR10353962 - 4,189 cells

Methods

The complete bioinformatics workflow consisted of several sequential stages. First, raw FASTQ files were evaluated using FastQC, followed by MultiQC for summarization of sequencing quality metrics.The reads were then processed using STARsolo for alignment, cell-barcode processing, UMI processing, and gene-level quantification. The resulting gene-by-cell count matrix was used for downstream analysis in Seurat.Seurat was used for quality control, cell filtering, normalization, identification of highly variable genes, principal component analysis (PCA), UMAP dimensionality reduction, clustering, marker gene identification, marker validation, and cell-type annotation.Cellular composition was subsequently compared between the three samples using absolute cell counts and relative cell proportions.A separate CNV analysis workflow was implemented using inferCNV. Candidate non-malignant populations were evaluated for use as reference cells. A balanced reference population of 300 cells was selected from Microglia-Macrophage and Oligodendrocyte populations. The reference matrix was validated for cell and gene identifier consistency, ordering, count validity, sample balance, and reference population composition. inferCNV was then used to investigate large-scale CNV-associated expression patterns. The final inferCNV results were validated against the original single-cell metadata.

Software and Packages

The analysis was performed using:

- R
- Seurat

- SeuratObject
- STARsolo
- FastQC
- MultiQC
- inferCNV
- Matrix
- ggplot2
- pheatmap
- dplyr
- Bioconductor
- GENCODE
- Linux / WSL2

Results

The single-cell analysis identified a heterogeneous cellular landscape within the glioblastoma samples.
Multiple transcriptionally distinct populations were identified, including:

- neural and glial populations
- oligodendrocyte-lineage populations
- immune populations
- endothelial populations
- mesenchymal populations
- cycling populations

The final cell-type annotation included populations such as Neural-like, Astrocyte-like, Oligodendrocyte, OPC-like, Microglia-Macrophage, Endothelial, Mesenchymal-like, and several cycling populations. The analysis of sample composition demonstrated differences in the relative abundance of cellular populations between the three samples. For the CNV analysis, a reference population of 300 cells was selected. The reference consisted of 236 Microglia-Macrophage cells and 64 Oligodendrocyte cells, with 100 reference cells selected from each sample. The inferCNV workflow included 13,632 cells, of which 300 were used as reference cells and 13,332 as observation cells. The resulting analysis was subsequently validated against the original metadata.

Discussion

The results demonstrate substantial cellular heterogeneity within the analyzed glioblastoma samples. The identification of multiple neural, glial, immune, endothelial, mesenchymal, and cycling populations illustrates the complex cellular composition of the GBM environment. Differences in cellular composition between samples indicate that the relative representation of individual populations varies between patients or biological samples. The Seurat workflow provides a comprehensive transcriptional characterization of the dataset, from initial quality control through clustering and cell-type annotation. The inferCNV analysis provides an exploratory framework for investigating large-scale CNV-associated expression patterns. Importantly, the current workflow does not classify cells as malignant or non-malignant solely on the basis of inferCNV signal.

Conclusions

This project demonstrates a complete single-cell RNA-seq analysis workflow applied to three glioblastoma samples.

The workflow successfully integrates:

- FASTQ quality control
- STARsolo alignment and quantification
- Seurat-based single-cell analysis
- PCA and UMAP
- clustering
- marker gene analysis
- cell-type annotation
- sample composition analysis
- reference population selection
- inferCNV analysis
- validation of CNV results

The analysis provides a reproducible framework for characterizing cellular heterogeneity in glioblastoma and for exploratory investigation of CNV-associated transcriptional patterns.

Repository Structure

```
03_SingleCell-RNAseq-Glioblastoma-Seurat/
│
├── README.md
├── docs/
│
├── data/
│   ├── counts/
│   ├── metadata/
│   ├── raw_fastq/
│   └── reference/
│
├── figures/
│   ├── QC_before_filtering.png
│   ├── QC_after_filtering.png
│   ├── QC_nCount_vs_nFeature_after.png
│   ├── PCA_ElbowPlot.png
│   ├── PCA_PC1_PC2_by_sample.png
│   ├── UMAP_clusters.png
│   ├── UMAP_by_sample.png
│   ├── UMAP_clusters_split_by_sample.png
│   ├── cluster_marker_heatmap.png
│   ├── CanonicalMarkers_DotPlot.png
│   ├── MarkerScores_by_cluster_heatmap.png
│   ├── UMAP_cell_type_annotation.png
│   ├── Cell_Composition_Counts.png
│   ├── Cell_Composition_Percent.png
│   ├── Cell_Composition_Heatmap.png
│   └── infercnv.png
│
├── results/
│
├── logs/
│
└── scripts/
    ├── 01_download_fastq.sh
    ├── 02_fastqc.sh
    ├── 03_multiqc.sh
    ├── 04_download_reference.sh
    ├── 05_download_star_index.sh
    ├── 06_download_whitelist.sh
    ├── 07_starsolo.sh
    ├── 08_seurat_qc.R
    ├── 09_seurat_filter.R
    ├── 10_seurat_normalization.R
    ├── 11_seurat_pca.R
    ├── 12_seurat_clustering_umap.R
    ├── 13_seurat_markers.R
    ├── 14_seurat_marker_validation.R
    ├── 15_seurat_annotation.R
    ├── 16_sample_composition.R
    ├── 17_prepare_cnv_annotation.R
    ├── 18_prepare_cnv_input.R
    ├── 19_validate_cnv_input.R
    ├── 20_prepare_cnv_counts.R
    ├── 21_validate_cnv_reference.R
    ├── 22_select_cnv_reference.R
    ├── 23_validate_selected_cnv_reference.R
    ├── 24_run_infercnv.R
    ├── 25_inspect_infercnv_results.R
    └── 26_validate_infercnv_observations.R
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